Handwritten Digit Recognition (MNIST)

MINI PROJECT REPORT

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श्रद्धावान् लभते ज्ञानम्

AMRITA SCHOOL OF COMPUTING, NAGERCOIL

AMRITA VISHWA VIDYAPEETHAM

NAGERCOIL 629901

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AMRITA VISHWA VIDYAPEETHAM
AMRITA SCHOOL OF COMPUTING, NAGERCOIL, 629901

श्रद्धावान् लभते ज्ञानम्

BONAFIDE CERTIFICATE

This is to certify that the mini project report titled **" Handwritten Digit Recognition (MNIST) "** submitted by **B VARSHITHA (Reg. No. NC.AI.U4AID24008)** is a bonafide record of the work carried out by the student. The work has been completed under my guidance and supervision at **Amrita School of Computing, Nagercoil**.

SIGNATURE

Faculty in-charge
Mrs. Anjana S
Assistant Professor/AID
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Nagercoil-629901

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ABSTRACT

Handwritten digit recognition is a fundamental problem in the field of computer vision and deep learning. This project presents a system for recognizing handwritten digits using a Convolutional Neural Network (CNN). The model is trained and tested using the MNIST dataset, which consists of grayscale images of digits ranging from 0 to 9.

The proposed system involves several stages, including data preprocessing, model building, training, and evaluation. The input images are normalized and reshaped to fit the requirements of the CNN model. The model consists of convolutional layers, pooling layers, and fully connected layers that help in extracting features and classifying the digits accurately.

The performance of the model is evaluated using accuracy metrics, along with visualization techniques such as accuracy and loss graphs, and a confusion matrix to analyze classification results. The model achieves an accuracy of approximately 98%, demonstrating its effectiveness in digit recognition tasks.

Additionally, the system supports prediction on custom input images, making it more practical and interactive. This project highlights the capability of deep learning techniques, especially CNNs, in solving real-world image classification problems efficiently.

CHAPTER-1

INTRODUCTION

Deep learning is a subfield of artificial intelligence that focuses on training neural networks to learn patterns from data. It has gained significant importance in recent years due to its ability to solve complex problems in areas such as image recognition, speech processing, and natural language understanding.

One of the important applications of deep learning is image classification, where a system is trained to identify and categorize images into different classes. Handwritten digit recognition is a classic example of image classification, where the goal is to correctly identify digits from 0 to 9 based on their visual representation.

Handwritten digit recognition plays a crucial role in many real-world applications such as postal mail sorting, bank cheque processing, form digitization, and automated data entry systems. Due to variations in handwriting styles, shapes, and sizes, it is a challenging problem to solve using traditional methods.

In this project, a Convolutional Neural Network (CNN) is used to recognize handwritten digits. CNNs are particularly effective for image-related tasks because they can automatically extract important features such as edges, patterns, and shapes from images. The model is trained using the MNIST dataset, which is a standard benchmark dataset in the field of machine learning.

The system is designed to preprocess the input data, train the CNN model, evaluate its performance, and predict digits from new images. The aim of this project is to develop an accurate and efficient model for handwritten digit recognition and to demonstrate the effectiveness of deep learning techniques in solving image classification problems.

CHAPTER-2

LITERATURE SURVEY

Handwritten digit recognition has been an important research area in the fields of machine learning and computer vision for several decades. Earlier approaches for digit recognition relied on traditional image processing techniques and simple machine learning algorithms such as k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and decision trees. These methods required manual feature extraction, where specific characteristics of digits, such as edges and shapes, were identified and used for classification.

With the advancement of artificial neural networks, more sophisticated models were developed to improve recognition accuracy. One of the most significant breakthroughs in this domain was the introduction of Convolutional Neural Networks (CNNs), which are capable of automatically learning features from images without the need for manual extraction. CNNs have shown superior performance in image classification tasks due to their ability to capture spatial hierarchies and patterns.

The MNIST dataset has been widely used as a benchmark for evaluating different algorithms in handwritten digit recognition. Several studies have demonstrated that CNN-based models achieve higher accuracy compared to traditional methods. Researchers have also explored techniques such as data augmentation, dropout, and batch normalization to further enhance model performance and reduce overfitting.

Recent advancements in deep learning frameworks have made it easier to implement and train complex models efficiently. Libraries such as TensorFlow and Keras provide powerful tools for building and deploying neural networks. These developments have significantly contributed to the widespread adoption of deep learning techniques in real-world applications.

This project builds upon these existing works by implementing a CNN-based model for handwritten digit recognition and evaluating its performance using standard metrics and visualization techniques.

CHAPTER-3

METHODOLOGY

The methodology of this project describes the step-by-step process followed to develop the handwritten digit recognition system using a Convolutional Neural Network (CNN). The overall process includes data collection, preprocessing, model design, training, and evaluation.

The dataset used in this project is the MNIST dataset, which contains 60,000 training images and 10,000 testing images of handwritten digits from 0 to 9. Each image is a grayscale image of size 28×28 pixels. This dataset is widely used as a benchmark for image classification problems.

In the preprocessing stage, the pixel values of the images are normalized to a range between 0 and 1 to improve model performance. The images are also reshaped to include a single channel, making them suitable for input into the CNN model.

The model architecture is based on a Convolutional Neural Network, which consists of multiple layers including convolutional layers, pooling layers, and fully connected layers. The convolutional layers are responsible for extracting important features from the images, while the pooling layers reduce the dimensionality and help in reducing computational complexity. The fully connected layers perform the final classification of the digits.

The model is trained using the training dataset with the Adam optimizer and sparse categorical cross-entropy loss function. During training, the model learns to identify patterns and features that distinguish different digits. The performance of the model is evaluated using the testing dataset to measure its accuracy.

Additionally, visualization techniques such as accuracy and loss graphs are used to monitor the training process. A confusion matrix is also generated to analyze the classification performance and identify any misclassifications.

This methodology ensures a systematic approach to building an efficient and accurate handwritten digit recognition system.

CHAPTER-4

IMPLEMENTATION

The implementation of the handwritten digit recognition system is carried out using Python programming language and deep learning libraries. The project is developed using TensorFlow and Keras, which provide high-level tools for building and training neural networks. The entire implementation is executed in the Google Colab environment, which offers cloud-based computing resources and easy access to datasets.

The first step in implementation involves loading the MNIST dataset using built-in functions available in Keras. The dataset is then preprocessed by normalizing the pixel values to a range between 0 and 1. This helps in improving the performance and convergence of the model during training. The images are reshaped into a format suitable for the CNN model, i.e., 28×28 pixels with a single channel.

Next, the Convolutional Neural Network model is constructed. The model consists of multiple layers, including convolutional layers with ReLU activation, max-pooling layers for dimensionality reduction, and fully connected dense layers for classification. The final output layer uses the softmax activation function to classify the input image into one of the ten digit classes.

After building the model, it is compiled using the Adam optimizer and sparse categorical cross-entropy loss function. The model is then trained using the training dataset for a

fixed number of epochs. During training, the model learns to identify patterns and features associated with different handwritten digits.

Once the training process is completed, the model is evaluated using the testing dataset to determine its accuracy. The results are visualized using graphs such as training and validation accuracy, and loss curves. A confusion matrix is also generated to analyze the classification performance.

In addition to this, the system allows prediction of custom input images. Users can upload their own handwritten digit images, which are preprocessed and passed to the trained model for prediction. Finally, the trained model is saved using the latest Keras format for future use without retraining.

This implementation demonstrates the practical application of deep learning techniques in solving real-world image classification problems effectively.

CHAPTER-5

RESULTS AND DISCUSSION

The performance of the handwritten digit recognition system is evaluated using the MNIST test dataset. After training the Convolutional Neural Network (CNN) model for a fixed number of epochs, the model achieves an accuracy of approximately 98%, indicating a high level of effectiveness in classifying handwritten digits.

The training and validation accuracy graphs show that the model learns efficiently over time, with accuracy increasing steadily across epochs. Similarly, the loss graphs demonstrate a decrease in both training and validation loss, indicating that the model is optimizing well and not suffering from significant overfitting.

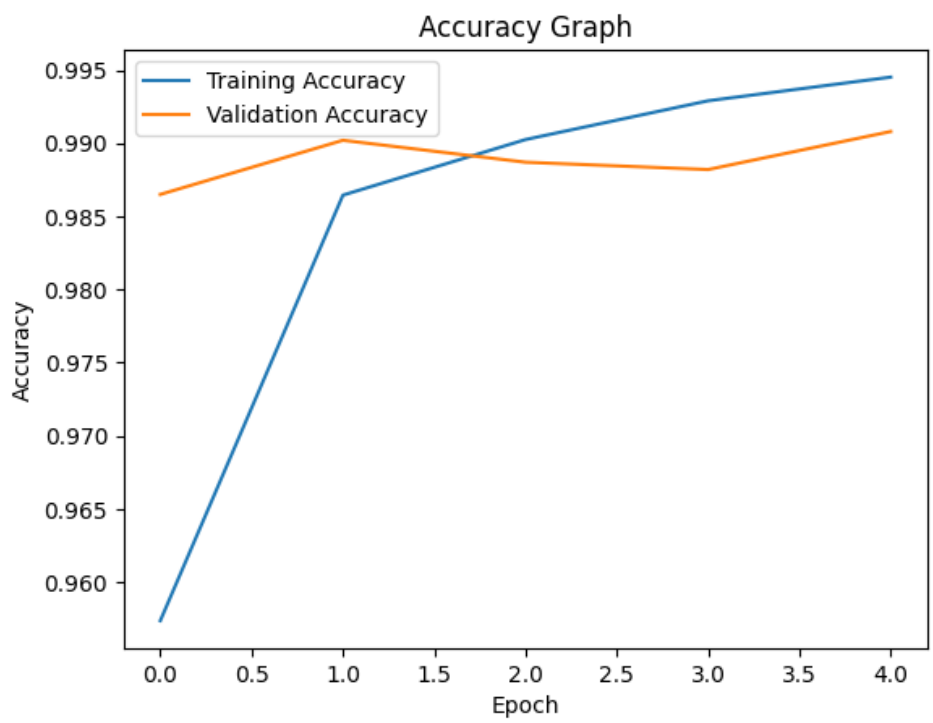
A confusion matrix is used to analyze the classification performance in detail. It provides a comparison between actual and predicted values, allowing identification of misclassified digits. The matrix shows that most predictions are correct, with only a few errors occurring between visually similar digits such as 3 and 5, or 4 and 9.

The model is also tested using sample images from the dataset, and it successfully predicts the correct digits in most cases. Additionally, the system supports custom image input, where users can upload their own handwritten digit images. The model processes these images and provides accurate predictions, demonstrating its practical usability.

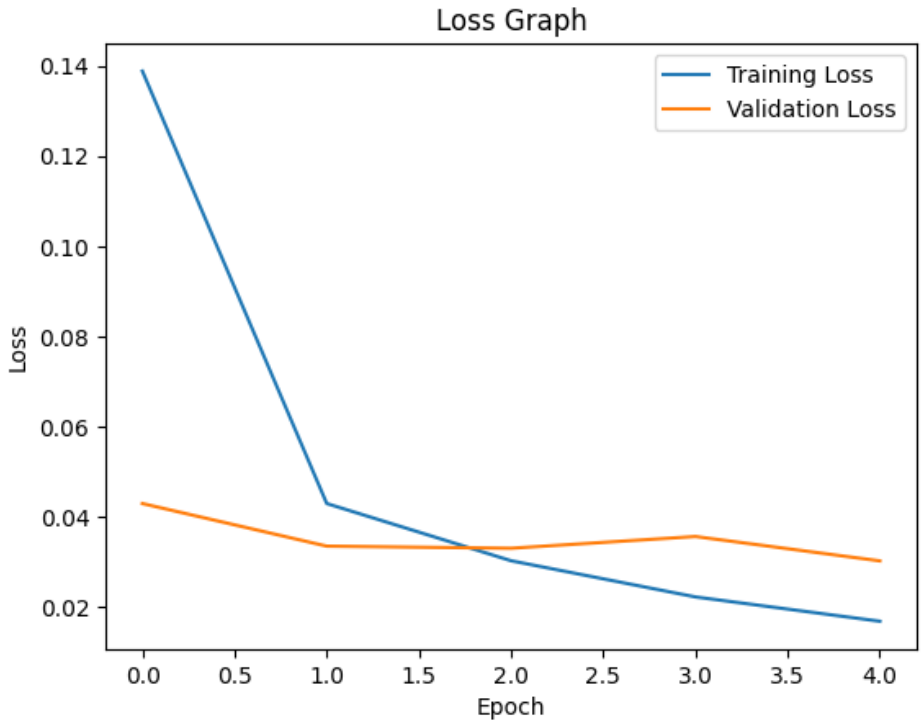
Overall, the results indicate that the CNN model performs efficiently and accurately for handwritten digit recognition. The combination of high accuracy, proper visualization, and real-time prediction capability makes the system reliable and effective for real-world applications.

<u>OUTPUT:</u>

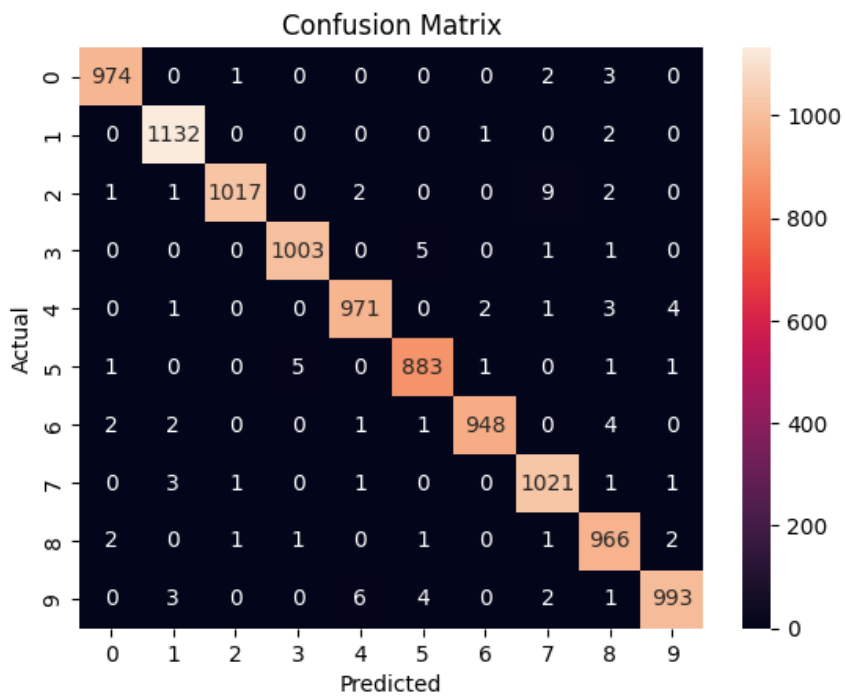
```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_sh
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 57s 30ms/step - accuracy: 0.9574 - loss: 0.1388 - val_accuracy: 0.9865 - val_loss: 0.0429
Epoch 2/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 55s 29ms/step - accuracy: 0.9865 - loss: 0.0429 - val_accuracy: 0.9902 - val_loss: 0.0335
Epoch 3/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 54s 29ms/step - accuracy: 0.9902 - loss: 0.0302 - val_accuracy: 0.9887 - val_loss: 0.0330
Epoch 4/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 54s 29ms/step - accuracy: 0.9929 - loss: 0.0222 - val_accuracy: 0.9882 - val_loss: 0.0356
Epoch 5/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 83s 29ms/step - accuracy: 0.9945 - loss: 0.0168 - val_accuracy: 0.9908 - val_loss: 0.0302
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 8ms/step - accuracy: 0.9908 - loss: 0.0302
```

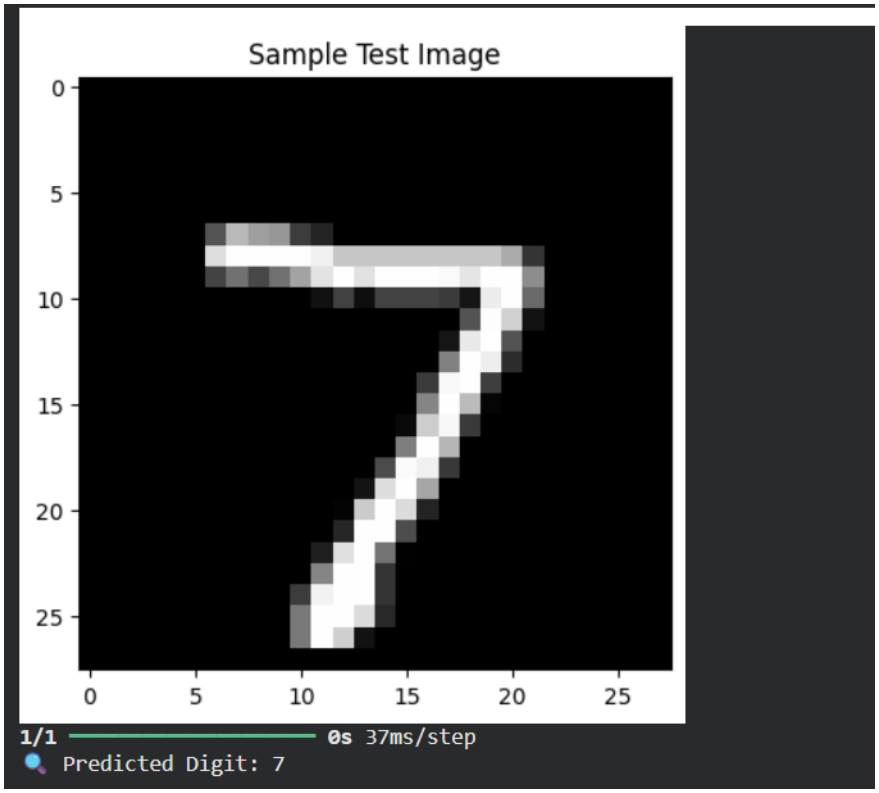


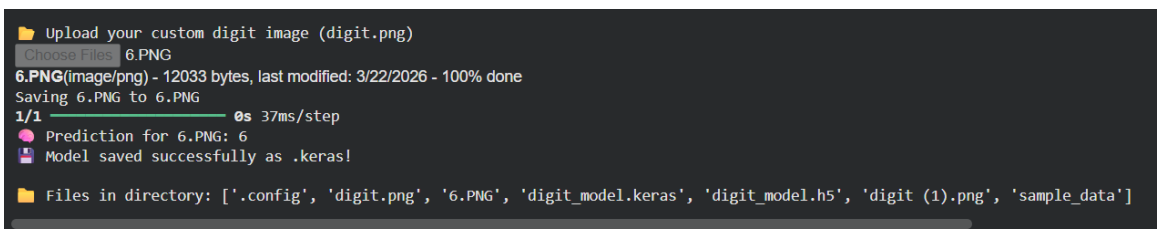

```
✅ Test Accuracy: 0.9908000230789185
```




```
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 8ms/step
```







CHAPTER-6

CONCLUSION

In this project, a handwritten digit recognition system was successfully developed using a Convolutional Neural Network (CNN). The model was trained and tested on the MNIST dataset, which is a standard benchmark dataset for image classification tasks. The system demonstrated the ability to accurately recognize handwritten digits from 0 to 9.

The preprocessing techniques such as normalization and reshaping played an important role in improving the performance of the model. The CNN architecture effectively extracted relevant features from the input images, enabling accurate classification. The model achieved an accuracy of approximately 98%, which indicates its reliability and efficiency.

The use of visualization techniques such as accuracy and loss graphs helped in understanding the training process, while the confusion matrix provided deeper insights into the model's performance and error patterns. The implementation of custom image prediction further enhanced the practicality of the system.

Overall, this project highlights the effectiveness of deep learning techniques, particularly CNNs, in solving image recognition problems. It also provides a strong foundation for further exploration in the field of artificial intelligence and computer vision.

CHAPTER-7

FUTURE SCOPE

Although the current system achieves high accuracy in recognizing handwritten digits, there are several areas where the project can be further improved and extended. One possible enhancement is to increase the complexity of the model by adding more layers or using advanced architectures to improve accuracy even further.
Another important improvement is the use of data augmentation techniques, which can help the model generalize better by exposing it to a wider variety of handwritten styles. This would make the system more robust when dealing with real-world inputs.
The project can also be extended to recognize alphabets and handwritten words instead of just digits. This would significantly increase its applicability in areas such as document digitization and automated form processing.
Additionally, the system can be deployed as a web or mobile application, allowing users to interact with it in real time. Integration with a camera for live digit recognition is another possible extension that would enhance usability.
Furthermore, optimization techniques can be applied to reduce model size and improve execution speed, making it suitable for deployment on low-resource devices.
Overall, the project has strong potential for expansion into more advanced and practical applications in the field of deep learning and computer vision.

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GITHUB LINK

https://github.com/varshispace/digit-recognition-project